



JasperGold® - Superlint Violation/Property Report

module *rv32i_core*created date *Feb 10 04:20:12 IST 2023*

VIOLATIONS SUMMARY

DESIGN_BUILD

#	Category	Error	Warning	Info	Waived	Total
1	BBOX	0	2	0	0	2

BBOX

#	Severity	Tag	Message	Source Location	Instance-Module	Grouping Status
1	Warning	VERI-9033	"array mem (size 32768) automatically BLACK-BOXED. Use the "elaborate -bbox_a" command to prevent automatic black-boxing."	inst_mem.v:11	*-inst_mem	-
2	Warning	VERI-9033	"array mem (size 8192) automatically BLACK-BOXED. Use the "elaborate -bbox_a" command to prevent automatic black-boxing."	data_memory.v:13	*-data_memory	-

Domain: LINT

#	Category	Error	Warning	Info	Waived	Total
1	CODINGSTYLE	0	0	0	140	140
2	STRUCTURAL	0	0	0	196	196
3	Total	0	0	0	336	336

CODINGSTYLE

#	Severity	Tag	Message	Source Location	Instance-Module	Grouping Status
1	Waived (ordering of bit should be decending order but in memory the depth i used in ascending order so it can be ignored)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit inst_mem -- should be all Descending ranges"	inst_mem.v:11	*-inst_mem	-
2	Waived (ordering of bit should be decending order but in memory the depth i used in ascending order so it can be ignored)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit register_file -- should be all Descending ranges"	register_file.v:17	*-register_file	-
3	Waived (ordering of bit should be decending order but in memory the depth i used in ascending order so it can be ignored)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit data_memory -- should be all Descending ranges"	data_memory.v:13	*-data_memory	-
4	Waived (PC arithmetic overflow is intentional due to fixed 32-bit PC	ASG_NR_POVF	"Potential overflow in assignment in design-unit/block 'btb'. LHS 'predict_target' (unsigned) is of 32 bit(s), RHS '(pc + 32'd4)' (unsigned) can be of 33 bit(s) "	btb.v:65	*-btb	-

	width; MSB truncation is architecturally valid.Arithmetic is modulo-XLEN by RISC-V spec. Overflow bits are intentionally discarded for PC, branch/jump targets, and ALU results)					
5	Waived (PC arithmetic overflow is intentional due to fixed 32-bit PC width; MSB truncation is architecturally valid.Arithmetic is modulo-XLEN by RISC-V spec. Overflow bits are intentionally discarded for PC, branch/jump targets, and ALU results)	ASG_NR_POVF	"Potential overflow in assignment in design-unit/block 'if_stage_simple_btb'. LHS 'pc_next' (unsigned) is of 32 bit(s), RHS '(pc_current + 32'd4)' (unsigned) can be of 33 bit(s) "	if_stage_simple_btb.v:110	*-if_stage_simple_btb	-
6	Waived (PC arithmetic overflow is intentional due to fixed 32-bit PC width; MSB truncation is architecturally valid.Arithmetic is modulo-XLEN by RISC-V spec. Overflow bits are intentionally discarded for PC, branch/jump targets, and ALU results)	ASG_NR_POVF	"Potential overflow in assignment in design-unit/block 'shift_unit32'. LHS 'result_shift' (unsigned) is of 32 bit(s), RHS '((rs1 >> shamt) ({32{rs1[31]}} << (6'd32 - {1'b0,shamt})))' (unsigned) can be of 64 bit(s) "	shift_unit32.v:90	*-shift_unit32	-
7	Waived (PC arithmetic overflow is intentional due to fixed 32-bit PC width; MSB truncation is architecturally valid.Arithmetic is modulo-XLEN by RISC-V spec. Overflow bits are intentionally discarded for PC, branch/jump targets, and ALU results)	ASG_NR_POVF	"Potential overflow in assignment in design-unit/block 'branch_jump_unit'. LHS 'target_branch_jal' (unsigned) is of 32 bit(s), RHS '(pc_ex + imm_ex)' (unsigned) can be of 33 bit(s) "	branch_jump_unit.v:71	*-branch_jump_unit	-
8	Waived (PC arithmetic overflow is intentional due to fixed 32-bit PC width; MSB truncation is architecturally valid.Arithmetic is	ASG_NR_POVF	"Potential overflow in assignment in design-unit/block 'branch_jump_unit'. LHS 'pc_plus_4' (unsigned) is of 32 bit(s), RHS '(pc_ex + 32'd4)' (unsigned) can be of 33 bit(s) "	branch_jump_unit.v:81	*-branch_jump_unit	-

	modulo-XLEN by RISC-V spec. Overflow bits are intentionally discarded for PC, branch/jump targets, and ALU results)					
9	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'state_n' assigned in case statement is not assigned to X in default case"	dynamic_branch_predictor.v:23	*-dynamic_branch_predictor	-
10	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'state_n' assigned in case statement is not assigned to X in default case"	dynamic_branch_predictor.v:31	*-dynamic_branch_predictor	-
11	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'imm_out' assigned in case statement is not assigned to X in default case"	decode_unit.v:26	*-decode_unit	-
12	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'alu_ctrl' assigned in case statement is not assigned to X in default case"	control_unit.v:32	*-control_unit	-
13	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'alu_ctrl' assigned in case statement is not assigned to X in default case"	control_unit.v:51	*-control_unit	-
14	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'mem_load_type' assigned in case statement is not assigned to X in default case"	control_unit.v:70	*-control_unit	-
15	Waived (default value should be unknown but	CAS_NR_DEFX	"Signal 'mem_store_type' assigned in case statement is not assigned to X in default case"	control_unit.v:87	*-control_unit	-

	default condition for my design should be all zeros not unknown values so this warning can be ignore)					
16	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'memtoreg' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
17	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'jal' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
18	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'alu_ctrl' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
19	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'mem_store_type' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
20	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'mem_load_type' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
21	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'ex_alu_src' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
22	Waived (default value should be unknown but default condition	CAS_NR_DEFX	"Signal 'wb_reg_file' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-

	for my design should be all zeros not unknown values so this warning can be ignore)					
23	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'mem_write' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
24	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'Branch_1' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
25	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'jalr' assigned in case statement is not assigned to X in default case"	control_unit.v:22	*-control_unit	-
26	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'result_alu' assigned in case statement is not assigned to X in default case"	logical_unit32.v:8	*-logical_unit32	-
27	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'result_cmp' assigned in case statement is not assigned to X in default case"	compare_unit32.v:8	*-compare_unit32	-
28	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'result_final' assigned in case statement is not assigned to X in default case"	alu_top32.v:43	*-alu_top32	-
29	Waived (default value should be unknown but default condition for my design	CAS_NR_DEFX	"Signal 'op1_sel' assigned in case statement is not assigned to X in default case"	execute_stage.v:39	*-execute_stage	-

	should be all zeros not unknown values so this warning can be ignore)					
30	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'op2_sel' assigned in case statement is not assigned to X in default case"	execute_stage.v:45	*-execute_stage	-
31	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'branch_cond' assigned in case statement is not assigned to X in default case"	branch_jump_unit.v:38	*-branch_jump_unit	-
32	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'byte_enable' assigned in case statement is not assigned to X in default case"	store_datapath.v:55	*-store_datapath	-
33	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'byte_enable' assigned in case statement is not assigned to X in default case"	store_datapath.v:49	*-store_datapath	-
34	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'mem_write_data' assigned in case statement is not assigned to X in default case"	store_datapath.v:49	*-store_datapath	-
35	Waived (default value should be unknown but default condition for my design should be all zeros not unknown values so this warning can be ignore)	CAS_NR_DEFX	"Signal 'read_data' assigned in case statement is not assigned to X in default case"	load_datapath.v:32	*-load_datapath	-
36	Waived (index should be 2 state data type but in memory index is address which is of wire data type)	IDX_NR_DTTY	"Variable 'wr_set' used as index in expression 'valid_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:82	*-btb_file	-

	that is 4 state data type so this warning can be ignore)					
37	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_way' used as index in expression 'valid_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:82	*-btb_file	-
38	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_set' used as index in expression 'tag_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:83	*-btb_file	-
39	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_way' used as index in expression 'tag_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:83	*-btb_file	-
40	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_set' used as index in expression 'target_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:84	*-btb_file	-
41	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_way' used as index in expression 'target_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:84	*-btb_file	-
42	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_set' used as index in expression 'state_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:85	*-btb_file	-
43	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state	IDX_NR_DTTY	"Variable 'wr_way' used as index in expression 'state_arr[wr_set][wr_way]' should be 2-state data type"	btb_file.v:85	*-btb_file	-

	data type so this warning can be ignore)					
44	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_set' used as index in expression 'lru[wr_set]' should be 2-state data type"	btb_file.v:89	*-btb_file	-
45	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'write_addr' used as index in expression 'mem[write_addr]' should be 2-state data type"	inst_mem.v:30	*-inst_mem	-
46	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'wr_addr' used as index in expression 'reg_file[wr_addr]' should be 2-state data type"	register_file.v:45	*-register_file	-
47	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'mem_addr' used as index in expression 'mem[mem_addr]' should be 2-state data type"	data_memory.v:31	*-data_memory	-
48	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'mem_addr' used as index in expression 'mem[(mem_addr + {{{(ADDR_BITS - 1){1'b0}},1'b1}})]' should be 2-state data type"	data_memory.v:32	*-data_memory	-
49	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this warning can be ignore)	IDX_NR_DTTY	"Variable 'mem_addr' used as index in expression 'mem[(mem_addr + {{{(ADDR_BITS - 2){1'b0}},2'b10}})]' should be 2-state data type"	data_memory.v:33	*-data_memory	-
50	Waived (index should be 2 state data type but in memory index is address which is of wire data type that is 4 state data type so this	IDX_NR_DTTY	"Variable 'mem_addr' used as index in expression 'mem[(mem_addr + {{{(ADDR_BITS - 2){1'b0}},2'b11}})]' should be 2-state data type"	data_memory.v:34	*-data_memory	-

	warning can be ignore)					
51	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_READ.hit0' is assigned asynchronously"	btb_read.v:14	u_if.u_btb.U_READ-btb_read -	
52	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_READ.hit1' is assigned asynchronously"	btb_read.v:15	u_if.u_btb.U_READ-btb_read -	

	be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
53	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_en' is assigned asynchronously"	btb_write.v:24	u_if.u_btb.U_WRITE-btb_write	-

	(Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
54	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_lru_en' is assigned asynchronously"	btb_write.v:33	u_if.u_btb.U_WRITE-btb_write	-

	cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
55	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_lru_val' is assigned asynchronously"	btb_write.v:34	u_if.u_btb.U_WRITE-btb_write	-

56	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_set' is assigned asynchronously"	btb_write.v:25	u_if.u_btb.U_WRITE-btb_write	-
57	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_state' is assigned asynchronously"	btb_write.v:30	u_if.u_btb.U_WRITE-btb_write	-

	flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
58	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_tag' is assigned asynchronously"	btb_write.v:28	u_if.u_btb.U_WRITE-btb_write	-

	path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
59	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_target' is assigned asynchronously"	btb_write.v:29	u_if.u_btb.U_WRITE-btb_write	-

	required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
60	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_valid' is assigned asynchronously"	btb_write.v:27	u_if.u_btb.U_WRITE-btb_write	-
61	Waived (output should be synchronous but	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_WRITE.wr_way' is assigned asynchronously"	btb_write.v:26	u_if.u_btb.U_WRITE-btb_write	-

	which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
62	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*,	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_lru' is assigned asynchronously"	btb_file.v:38	u_if.u_btb.U_FILE-btb_file	-

	rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
63	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_state0' is assigned asynchronously"	btb_file.v:20	u_if.u_btb.U_FILE-btb_file	-

	stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
64	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_state1' is assigned asynchronously"	btb_file.v:26	u_if.u_btb.U_FILE-btb_file	-

	behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
65	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_tag0' is assigned asynchronously"	btb_file.v:18	u_if.u_btb.U_FILE-btb_file	-
66	Waived (output should be synchronous but which modules does not have clock, having this	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_tag1' is assigned asynchronously"	btb_file.v:24	u_if.u_btb.U_FILE-btb_file	-

	warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
67	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid,	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_target0' is assigned asynchronously"	btb_file.v:19	u_if.u_btb.U_FILE-btb_file	-

	predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
68	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_target1' is assigned asynchronously"	btb_file.v:25	u_if.u_btb.U_FILE-btb_file	-

	(hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
69	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct,	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_valid0' is assigned asynchronously"	btb_file.v:17	u_if.u_btb.U_FILE-btb_file	-

	synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
70	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_if.u_btb.U_FILE.rd_valid1' is assigned asynchronously"	btb_file.v:23	u_if.u_btb.U_FILE-btb_file	-
71	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having	OTP_NR_ASYA	"Output port 'u_if.u_btb.predict_taken' is assigned asynchronously"	btb.v:12	u_if.u_btb-btb	-

	clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
72	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally	OTP_NR_ASYA	"Output port 'u_if.u_btb.predict_target' is assigned asynchronously"	btb.v:13	u_if.u_btb-btb	-

	driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
73	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the	OTP_NR_ASYA	"Output port 'u_if.u_btb.predict_valid' is assigned asynchronously"	btb.v:11	u_if.u_btb-btb	-

	PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
74	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are	OTP_NR_ASYA	"Output port 'u_if.u_imem.instruction' is assigned asynchronously"	inst_mem.v:9	u_if.u_imem-inst_mem	-

	intentional, safe, and required by design.)					
75	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_if.instr_if' is assigned asynchronously"	if_stage_simple_btb.v:23	u_if-if_stage_simple_btb	-
76	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so	OTP_NR_ASYA	"Output port 'u_if.predictedTaken_if' is assigned asynchronously"	if_stage_simple_btb.v:24	u_if-if_stage_simple_btb	-

	this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
77	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a	OTP_NR_ASYA	"Output port 'u_decode.u_decode_unit.imm_out' is assigned asynchronously"	decode_unit.v:6	u_decode.u_decode_unit-decode_unit	-

	zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
78	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.Branch_1' is assigned asynchronously"	control_unit.v:13	u_decode.u_ctrl-control_unit -	

	an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
79	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.alu_ctrl' is assigned asynchronously"	control_unit.v:18	u_decode.u_ctrl-control_unit -	

80	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.ex_alu_src' is assigned asynchronously"	control_unit.v:6	u_decode.u_ctrl-control_unit -	
81	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.jal' is assigned asynchronously"	control_unit.v:14	u_decode.u_ctrl-control_unit -	

	flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
82	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.jalr' is assigned asynchronously"	control_unit.v:15	u_decode.u_ctrl-control_unit -	

	path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
83	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.mem_load_type' is assigned asynchronously"	control_unit.v:9	u_decode.u_ctrl-control_unit -	

	required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
84	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.mem_store_type' is assigned asynchronously"	control_unit.v:10	u_decode.u_ctrl-control_unit -	
85	Waived (output should be synchronous but	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.mem_write' is assigned asynchronously"	control_unit.v:7	u_decode.u_ctrl-control_unit -	

	which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
86	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*,	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.memtoreg' is assigned asynchronously"	control_unit.v:12	u_decode.u_ctrl-control_unit -	

	rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
87	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch	OTP_NR_ASYA	"Output port 'u_decode.u_ctrl.wb_reg_file' is assigned asynchronously"	control_unit.v:11	u_decode.u_ctrl-control_unit -	

	stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
88	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This	OTP_NR_ASYA	"Output port 'u_decode.u_regfile.rs1_data' is assigned asynchronously"	register_file.v :14	u_decode.u_regfile-register_file	-

	behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
89	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_decode.u_regfile.rs2_data' is assigned asynchronously"	register_file.v:15	u_decode.u_regfile-register_file	-
90	Waived (output should be synchronous but which modules does not have clock, having this	OTP_NR_ASYA	"Output port 'u_decode.Branch_1' is assigned asynchronously"	top_decode.v:36	u_decode-top_decode	-

	warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
91	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid,	OTP_NR_ASYA	"Output port 'u_decode.alu_ctrl' is assigned asynchronously"	top_decode.v :41	u_decode-top_decode	-

	predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
92	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results	OTP_NR_ASYA	"Output port 'u_decode.ex_alu_src' is assigned asynchronously"	top_decode.v :29	u_decode-top_decode	-

	(hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
93	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct,	OTP_NR_ASYA	"Output port 'u_decode.imm_out' is assigned asynchronously"	top_decode.v:22	u_decode-top_decode	-

	synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
94	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_decode.jal' is assigned asynchronously"	top_decode.v :37	u_decode-top_decode	-
95	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having	OTP_NR_ASYA	"Output port 'u_decode.jalr' is assigned asynchronously"	top_decode.v :38	u_decode-top_decode	-

	clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
96	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally	OTP_NR_ASYA	"Output port 'u_decode.mem_load_type' is assigned asynchronously"	top_decode.v:32	u_decode-top_decode	-

	driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
97	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the	OTP_NR_ASYA	"Output port 'u_decode.mem_store_type' is assigned asynchronously"	top_decode.v :33	u_decode-top_decode	-

	PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
98	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are	OTP_NR_ASYA	"Output port 'u_decode.mem_write' is assigned asynchronously"	top_decode.v :30	u_decode-top_decode	-

	intentional, safe, and required by design.)					
99	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_decode.memtoreg' is assigned asynchronously"	top_decode.v :35	u_decode-top_decode	-
100	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so	OTP_NR_ASYA	"Output port 'u_decode.rs1_data' is assigned asynchronously"	top_decode.v :25	u_decode-top_decode	-

	this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
101	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a	OTP_NR_ASYA	"Output port 'u_decode.rs2_data' is assigned asynchronously"	top_decode.v :26	u_decode-top_decode	-

	zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
102	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce	OTP_NR_ASYA	"Output port 'u_decode.wb_reg_file' is assigned asynchronously"	top_decode.v :34	u_decode-top_decode	-

	an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
103	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_hazard.id_ex_flush' is assigned asynchronously"	hazard_unit.v:11	u_hazard-hazard_unit	-

104	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_hazard.if_id_flush' is assigned asynchronously"	hazard_unit.v:10	u_hazard-hazard_unit	-
105	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The	OTP_NR_ASYA	"Output port 'u_hazard.pc_en' is assigned asynchronously"	hazard_unit.v:8	u_hazard-hazard_unit	-

	flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
106	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch	OTP_NR_ASYA	"Output port 'u_fwd.operand_a_forward_cntl' is assigned asynchronously"	forwarding_unit.v:9	u_fwd-forwarding_unit	-

	path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
107	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the	OTP_NR_ASYA	"Output port 'u_fwd.operand_b_forward_cntl' is assigned asynchronously"	forwarding_unit.v:10	u_fwd-forwarding_unit	-

	required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
108	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_arith.carry_flag' is assigned asynchronously"	arithmetic_unit32.v:71	u_exe.u_alu_top.u_arith-arithmetic_unit32	-
109	Waived (output should be synchronous but	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_arith.negative_flag' is assigned asynchronously"	arithmetic_unit32.v:72	u_exe.u_alu_top.u_arith-arithmetic_unit32	-

	which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
110	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*,	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_arith.overflow_flag' is assigned asynchronously"	arithmetic_unit32.v:73	u_exe.u_alu_top.u_arith-arithmetic_unit32	-

	rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
111	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_arith.result_alu' is assigned asynchronously"	arithmetic_unit32.v:69	u_exe.u_alu_top.u_arith- arithmetic_unit32	-

	stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
112	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_arith.zero_flag' is assigned asynchronously"	arithmetic_unit32.v:70	u_exe.u_alu_top.u_arith-arithmetic_unit32	-

	behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
113	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_logic.result_alu' is assigned asynchronously"	logical_unit32.v:5	u_exe.u_alu_top.u_logic-logical_unit32	-
114	Waived (output should be synchronous but which modules does not have clock, having this	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_shift.result_shift' is assigned asynchronously"	shift_unit32.v:72	u_exe.u_alu_top.u_shift-shift_unit32	-

	warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
115	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid,	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.u_cmp.result_cmp' is assigned asynchronously"	compare_unit32.v:5	u_exe.u_alu_top.u_cmp-compare_unit32	-

	predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
116	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.alu_result' is assigned asynchronously"	alu_top32.v:8	u_exe.u_alu_top-alu_top32	-

	(hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)				
117	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct,	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.carry_flag' is assigned asynchronously"	alu_top32.v:11	u_exe.u_alu_top-alu_top32 -

	synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
118	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.negative_flag' is assigned asynchronously"	alu_top32.v:10	u_exe.u_alu_top-alu_top32	-
119	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.overflow_flag' is assigned asynchronously"	alu_top32.v:12	u_exe.u_alu_top-alu_top32	-

	clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
120	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally	OTP_NR_ASYA	"Output port 'u_exe.u_alu_top.zero_flag' is assigned asynchronously"	alu_top32.v:9	u_exe.u_alu_top-alu_top32	-

	driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
121	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the	OTP_NR_ASYA	"Output port 'u_exe.alu_result_ex' is assigned asynchronously"	execute_stage.v:16	u_exe-execute_stage	-

	PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
122	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are	OTP_NR_ASYA	"Output port 'u_exe.carry_flag_ex' is assigned asynchronously"	execute_stage.v:19	u_exe-execute_stage	-

	intentional, safe, and required by design.)					
123	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_exe.negative_flag_ex' is assigned asynchronously"	execute_stage.v:18	u_exe-execute_stage	-
124	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so	OTP_NR_ASYA	"Output port 'u_exe.op1_selected_ex' is assigned asynchronously"	execute_stage.v:23	u_exe-execute_stage	-

	this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
125	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a	OTP_NR_ASYA	"Output port 'u_exe.overflow_flag_ex' is assigned asynchronously"	execute_stage.v:20	u_exe-execute_stage	-

	zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
126	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce	OTP_NR_ASYA	"Output port 'u_exe.rs2_data_for_mem_ex' is assigned asynchronously"	execute_stage.v:21	u_exe-execute_stage	-

	an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
127	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_exe.zero_flag_ex' is assigned asynchronously"	execute_stage.v:17	u_exe-execute_stage	-

128	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_branch.ex_branch_taken' is assigned asynchronously"	branch_jump_unit.v:18	u_branch-branch_jump_unit	-
129	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The	OTP_NR_ASYA	"Output port 'u_branch.jump_addr_ex' is assigned asynchronously"	branch_jump_unit.v:21	u_branch-branch_jump_unit	-

	flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
130	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch	OTP_NR_ASYA	"Output port 'u_branch.modify_pc_ex' is assigned asynchronously"	branch_jump_unit.v:19	u_branch-branch_jump_unit	-

	path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
131	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the	OTP_NR_ASYA	"Output port 'u_branch.update_btb_ex' is assigned asynchronously"	branch_jump_unit.v:22	u_branch-branch_jump_unit	-

	required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
132	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_branch.update_pc_ex' is assigned asynchronously"	branch_jump_unit.v:20	u_branch-branch_jump_unit	-
133	Waived (output should be synchronous but	OTP_NR_ASYA	"Output port 'u_mem.u_datamem.u_store.byte_enable' is assigned asynchronously"	store_datapath.v:42	u_mem.u_datamem.u_store-store_datapath	-

	which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
134	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*,	OTP_NR_ASYA	"Output port 'u_mem.u_datamem.u_store.mem_write_data' is assigned asynchronously"	store_datapath.v:41	u_mem.u_datamem.u_store-store_datapath	-

	rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
135	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch	OTP_NR_ASYA	"Output port 'u_mem.u_datamem.u_mem.mem_data_out' is assigned asynchronously"	data_memory.v:8	u_mem.u_datamem.u_mem- data_memory	-

	stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
136	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*, predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This	OTP_NR_ASYA	"Output port 'u_mem.u_datamem.u_load.read_data' is assigned asynchronously"	load_datapath.v:10	u_mem.u_datamem.u_load-load_datapath	-

	behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
137	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)	OTP_NR_ASYA	"Output port 'u_mem.u_datamem.read_data' is assigned asynchronously"	data_mem_top.v:9	u_mem.u_datamem-data_mem_top	-
138	Waived (output should be synchronous but which modules does not have clock, having this	OTP_NR_ASYA	"Output port 'u_mem.load_wb_data' is assigned asynchronously"	mem_stage.v:88	u_mem-mem_stage	-

	warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
139	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid,	OTP_NR_ASYA	"Output port 'u_wb.wb_write_data' is assigned asynchronously"	wb_stage.v:12	u_wb-wb_stage	-

	predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results (hit, taken, target) in the same cycle as the PC.Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing.This behavior is architecturally correct, synthesizable The asynchronous assignments are intentional, safe, and required by design.)					
140	Waived (output should be synchronous but which modules does not have clock, having this warning in output port and which module having clock, reading data is asynchronous. so this warning can be ignore . For BTB design The flagged output ports (rd_valid*, rd_tag*, rd_target*, rd_state*, hit*,predict_valid, predict_taken, predict_target) are intentionally driven by combinational logic as part of a zero-cycle BTB (Branch Target Buffer) fetch path. The BTB is accessed in the instruction fetch stage and must produce branch prediction results	OTP_NR_ASYA	"Output port 'u_wb.wb_write_en' is assigned asynchronously"	wb_stage.v:14	u_wb-wb_stage	-

(hit, taken, target) in the same cycle as the PC. Registering these outputs would introduce an additional cycle of latency and violate the required branch prediction timing. This behavior is architecturally correct, synthesizable. The asynchronous assignments are intentional, safe, and required by design.)					
---	--	--	--	--	--

STRUCTURAL

#	Severity	Tag	Message	Source Location	Instance-Module	Grouping Status
1	Waived (Register file flip-flops intentionally omit asynchronous reset. Register contents are architecturally undefined after reset and initialized by software. This reduces area and reset fanout.)	FLP_NO_ASRT	"Flip-flop 'reg_file' does not have any asynchronous set or reset"	register_file.v:43	*-register_file	-
2	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_pc_reg.next_pc' of instance 'u_if.u_pc_reg' is not a register"	pc_reg.v:6	u_if.u_pc_reg-pc_reg	-
3	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_pc_reg.pc_en' of instance 'u_if.u_pc_reg' is not a register"	pc_reg.v:5	u_if.u_pc_reg-pc_reg	-
4	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_READ.pc' of instance 'u_if.u_btb.U_READ' is not a register"	btb_read.v:5	u_if.u_btb.U_READ-btb_read	-
5	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_READ.rd_tag0' of instance 'u_if.u_btb.U_READ' is not a register"	btb_read.v:9	u_if.u_btb.U_READ-btb_read	-
6	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_READ.rd_tag1' of instance 'u_if.u_btb.U_READ' is not a register"	btb_read.v:11	u_if.u_btb.U_READ-btb_read	-
7	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_READ.rd_valid0' of instance 'u_if.u_btb.U_READ' is not a register"	btb_read.v:8	u_if.u_btb.U_READ-btb_read	-
8	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_READ.rd_valid1' of instance 'u_if.u_btb.U_READ' is not a register"	btb_read.v:10	u_if.u_btb.U_READ-btb_read	-

9	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.dp0.actual_taken' of instance 'u_if.u_btb.U_WRITE.dp0' is not a register"	dynamic_branch_predictor.v:12	u_if.u_btb.U_WRITE.dp0-dynamic_branch_predictor	-
10	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.dp0.curr_state' of instance 'u_if.u_btb.U_WRITE.dp0' is not a register"	dynamic_branch_predictor.v:11	u_if.u_btb.U_WRITE.dp0-dynamic_branch_predictor	-
11	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.dp0.update_en' of instance 'u_if.u_btb.U_WRITE.dp0' is not a register"	dynamic_branch_predictor.v:9	u_if.u_btb.U_WRITE.dp0-dynamic_branch_predictor	-
12	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.dp1.actual_taken' of instance 'u_if.u_btb.U_WRITE.dp1' is not a register"	dynamic_branch_predictor.v:12	u_if.u_btb.U_WRITE.dp1-dynamic_branch_predictor	-
13	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.dp1.curr_state' of instance 'u_if.u_btb.U_WRITE.dp1' is not a register"	dynamic_branch_predictor.v:11	u_if.u_btb.U_WRITE.dp1-dynamic_branch_predictor	-
14	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.dp1.update_en' of instance 'u_if.u_btb.U_WRITE.dp1' is not a register"	dynamic_branch_predictor.v:9	u_if.u_btb.U_WRITE.dp1-dynamic_branch_predictor	-
15	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.actual_taken' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:11	u_if.u_btb.U_WRITE-btb_write	-
16	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.rd_lru_upd' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:21	u_if.u_btb.U_WRITE-btb_write	-
17	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.rd_tag0_upd' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:17	u_if.u_btb.U_WRITE-btb_write	-
18	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.rd_tag1_upd' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:20	u_if.u_btb.U_WRITE-btb_write	-
19	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.rd_valid0_upd' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:16	u_if.u_btb.U_WRITE-btb_write	-
20	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.rd_valid1_upd' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:19	u_if.u_btb.U_WRITE-btb_write	-
21	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.state0_in' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:37	u_if.u_btb.U_WRITE-btb_write	-
22	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.state1_in' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:38	u_if.u_btb.U_WRITE-btb_write	-

	this warning can ignore)					
23	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.update_en' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:9	u_if.u_btb.U_WRITE-btb_write	-
24	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.update_pc' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:10	u_if.u_btb.U_WRITE-btb_write	-
25	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_WRITE.update_target' of instance 'u_if.u_btb.U_WRITE' is not a register"	btb_write.v:12	u_if.u_btb.U_WRITE-btb_write	-
26	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.rd_set' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:14	u_if.u_btb.U_FILE-btb_file	-
27	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_en' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:29	u_if.u_btb.U_FILE-btb_file	-
28	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_lru_en' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:39	u_if.u_btb.U_FILE-btb_file	-
29	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_lru_val' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:40	u_if.u_btb.U_FILE-btb_file	-
30	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_set' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:30	u_if.u_btb.U_FILE-btb_file	-
31	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_state' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:35	u_if.u_btb.U_FILE-btb_file	-
32	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_tag' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:33	u_if.u_btb.U_FILE-btb_file	-
33	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_target' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:34	u_if.u_btb.U_FILE-btb_file	-
34	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_valid' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:32	u_if.u_btb.U_FILE-btb_file	-
35	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.U_FILE.wr_way' of instance 'u_if.u_btb.U_FILE' is not a register"	btb_file.v:31	u_if.u_btb.U_FILE-btb_file	-
36	Waived (input should be register, but some inputs is taken from the	INS_NR_INPR	"Input port 'u_if.u_btb.actual_taken' of instance 'u_if.u_btb' is not a register"	btb.v:18	u_if.u_btb-btb	-

	combinational module so this warning can ignore)					
37	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.pc' of instance 'u_if.u_btb' is not a register"	btb.v:10	u_if.u_btb-btb	-
38	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.update_en' of instance 'u_if.u_btb' is not a register"	btb.v:16	u_if.u_btb-btb	-
39	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.update_pc' of instance 'u_if.u_btb' is not a register"	btb.v:17	u_if.u_btb-btb	-
40	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_btb.update_target' of instance 'u_if.u_btb' is not a register"	btb.v:19	u_if.u_btb-btb	-
41	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_imem.clk' of instance 'u_if.u_imem' is not a register"	inst_mem.v:2	u_if.u_imem-inst_mem	-
42	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_imem.pc' of instance 'u_if.u_imem' is not a register"	inst_mem.v:4	u_if.u_imem-inst_mem	-
43	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_imem.rst' of instance 'u_if.u_imem' is not a register"	inst_mem.v:3	u_if.u_imem-inst_mem	-
44	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_imem.write_addr' of instance 'u_if.u_imem' is not a register"	inst_mem.v:6	u_if.u_imem-inst_mem	-
45	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_imem.write_data' of instance 'u_if.u_imem' is not a register"	inst_mem.v:7	u_if.u_imem-inst_mem	-
46	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.u_imem.write_en' of instance 'u_if.u_imem' is not a register"	inst_mem.v:5	u_if.u_imem-inst_mem	-
47	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.ex_branch_taken' of instance 'u_if' is not a register"	if_stage_simple_btb.v:19	u_if-if_stage_simple_btb	-
48	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.jump_addr_ex' of instance 'u_if' is not a register"	if_stage_simple_btb.v:17	u_if-if_stage_simple_btb	-
49	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.modify_pc_ex' of instance 'u_if' is not a register"	if_stage_simple_btb.v:14	u_if-if_stage_simple_btb	-

50	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.pc_en' of instance 'u_if' is not a register"	if_stage_simple_btb.v:10	u_if-if_stage_simple_btb	-
51	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.pc_ex' of instance 'u_if' is not a register"	if_stage_simple_btb.v:16	u_if-if_stage_simple_btb	-
52	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.rst_im' of instance 'u_if' is not a register"	if_stage_simple_btb.v:4	u_if-if_stage_simple_btb	-
53	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.update_btb_ex' of instance 'u_if' is not a register"	if_stage_simple_btb.v:18	u_if-if_stage_simple_btb	-
54	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.update_pc_ex' of instance 'u_if' is not a register"	if_stage_simple_btb.v:15	u_if-if_stage_simple_btb	-
55	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.write_addr' of instance 'u_if' is not a register"	if_stage_simple_btb.v:6	u_if-if_stage_simple_btb	-
56	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.write_data' of instance 'u_if' is not a register"	if_stage_simple_btb.v:7	u_if-if_stage_simple_btb	-
57	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if.write_en' of instance 'u_if' is not a register"	if_stage_simple_btb.v:5	u_if-if_stage_simple_btb	-
58	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if_id.en' of instance 'u_if_id' is not a register"	if_id_pipe.v:5	u_if_id-if_id_pipe	-
59	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if_id.flush' of instance 'u_if_id' is not a register"	if_id_pipe.v:6	u_if_id-if_id_pipe	-
60	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if_id.instr_in' of instance 'u_if_id' is not a register"	if_id_pipe.v:8	u_if_id-if_id_pipe	-
61	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if_id.pc_in' of instance 'u_if_id' is not a register"	if_id_pipe.v:7	u_if_id-if_id_pipe	-
62	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_if_id.predictedTaken_in' of instance 'u_if_id' is not a register"	if_id_pipe.v:9	u_if_id-if_id_pipe	-
63	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_decode_unit.id_flush' of instance 'u_decode.u_decode_unit' is not a register"	decode_unit.v:4	u_decode.u_decode_unit-decode_unit	-

	this warning can ignore)					
64	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_decode_unit.instruction_in' of instance 'u_decode.u_decode_unit' is not a register"	decode_unit.v:2	u_decode.u_decode_unit-decode_unit	-
65	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_ctrl.func3' of instance 'u_decode.u_ctrl' is not a register"	control_unit.v:3	u_decode.u_ctrl-control_unit	-
66	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_ctrl.func7' of instance 'u_decode.u_ctrl' is not a register"	control_unit.v:4	u_decode.u_ctrl-control_unit	-
67	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_ctrl.opcode' of instance 'u_decode.u_ctrl' is not a register"	control_unit.v:2	u_decode.u_ctrl-control_unit	-
68	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_regfile.rs1_addr' of instance 'u_decode.u_regfile' is not a register"	register_file.v:10	u_decode.u_regfile-register_file	-
69	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_regfile.rs2_addr' of instance 'u_decode.u_regfile' is not a register"	register_file.v:11	u_decode.u_regfile-register_file	-
70	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_regfile.wr_addr' of instance 'u_decode.u_regfile' is not a register"	register_file.v:6	u_decode.u_regfile-register_file	-
71	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_regfile.wr_data' of instance 'u_decode.u_regfile' is not a register"	register_file.v:7	u_decode.u_regfile-register_file	-
72	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.u_regfile.wr_en' of instance 'u_decode.u_regfile' is not a register"	register_file.v:5	u_decode.u_regfile-register_file	-
73	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.id_flush' of instance 'u_decode' is not a register"	top_decode.v:8	u_decode-top_decode	-
74	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.instruction_in' of instance 'u_decode' is not a register"	top_decode.v:7	u_decode-top_decode	-
75	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.wb_wr_addr' of instance 'u_decode' is not a register"	top_decode.v:12	u_decode-top_decode	-
76	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_decode.wb_wr_data' of instance 'u_decode' is not a register"	top_decode.v:13	u_decode-top_decode	-
77	Waived (input should be register, but some inputs is taken from the	INS_NR_INPR	"Input port 'u_decode.wb_wr_en' of instance 'u_decode' is not a register"	top_decode.v:11	u_decode-top_decode	-

	combinational module so this warning can ignore)					
78	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_hazard.ex_load_inst' of instance 'u_hazard' is not a register"	hazard_unit.v:6	u_hazard-hazard_unit	-
79	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_hazard.ex_rd' of instance 'u_hazard' is not a register"	hazard_unit.v:5	u_hazard-hazard_unit	-
80	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_hazard.id_rs1' of instance 'u_hazard' is not a register"	hazard_unit.v:2	u_hazard-hazard_unit	-
81	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_hazard.id_rs2' of instance 'u_hazard' is not a register"	hazard_unit.v:3	u_hazard-hazard_unit	-
82	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_hazard.modify_pc_ex' of instance 'u_hazard' is not a register"	hazard_unit.v:7	u_hazard-hazard_unit	-
83	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_hazard.opcode_id' of instance 'u_hazard' is not a register"	hazard_unit.v:4	u_hazard-hazard_unit	-
84	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.Branch_1' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:21	u_id_ex-id_ex_pipe	-
85	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.alu_ctrl' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:24	u_id_ex-id_ex_pipe	-
86	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.en' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:4	u_id_ex-id_ex_pipe	-
87	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.ex_alu_src' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:15	u_id_ex-id_ex_pipe	-
88	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.flush' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:5	u_id_ex-id_ex_pipe	-
89	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.func3' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:8	u_id_ex-id_ex_pipe	-
90	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.imm_out' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:12	u_id_ex-id_ex_pipe	-

91	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.jal' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:22	u_id_ex-id_ex_pipe	-
92	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.jalr' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:23	u_id_ex-id_ex_pipe	-
93	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.mem_load_type' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:17	u_id_ex-id_ex_pipe	-
94	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.mem_store_type' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:18	u_id_ex-id_ex_pipe	-
95	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.mem_write' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:16	u_id_ex-id_ex_pipe	-
96	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.memtoreg' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:20	u_id_ex-id_ex_pipe	-
97	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.pc_id' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:6	u_id_ex-id_ex_pipe	-
98	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.predictedTaken_id' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:7	u_id_ex-id_ex_pipe	-
99	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.rd' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:9	u_id_ex-id_ex_pipe	-
100	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.rs1' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:10	u_id_ex-id_ex_pipe	-
101	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.rs1_data' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:13	u_id_ex-id_ex_pipe	-
102	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.rs2' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:11	u_id_ex-id_ex_pipe	-
103	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.rs2_data' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:14	u_id_ex-id_ex_pipe	-
104	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_id_ex.wb_reg_file' of instance 'u_id_ex' is not a register"	id_ex_pipe.v:19	u_id_ex-id_ex_pipe	-

	this warning can ignore)					
105	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_fwd.exmem_rd' of instance 'u_fwd' is not a register"	forwarding_unit.v:6	u_fwd-forwarding_unit	-
106	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_fwd.exmem_regwrite' of instance 'u_fwd' is not a register"	forwarding_unit.v:5	u_fwd-forwarding_unit	-
107	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_fwd.memwb_rd' of instance 'u_fwd' is not a register"	forwarding_unit.v:8	u_fwd-forwarding_unit	-
108	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_fwd.memwb_regwrite' of instance 'u_fwd' is not a register"	forwarding_unit.v:7	u_fwd-forwarding_unit	-
109	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_fwd.rs1_ex' of instance 'u_fwd' is not a register"	forwarding_unit.v:3	u_fwd-forwarding_unit	-
110	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_fwd.rs2_ex' of instance 'u_fwd' is not a register"	forwarding_unit.v:4	u_fwd-forwarding_unit	-
111	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_arith.alu_ctrl' of instance 'u_exe.u_alu_top.u_arith' is not a register"	arithmetic_unit32.v:67	u_exe.u_alu_top.u_arith-arithmetic_unit32	-
112	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_arith.rs1' of instance 'u_exe.u_alu_top.u_arith' is not a register"	arithmetic_unit32.v:65	u_exe.u_alu_top.u_arith-arithmetic_unit32	-
113	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_arith.rs2' of instance 'u_exe.u_alu_top.u_arith' is not a register"	arithmetic_unit32.v:66	u_exe.u_alu_top.u_arith-arithmetic_unit32	-
114	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_logic.alu_ctrl' of instance 'u_exe.u_alu_top.u_logic' is not a register"	logical_unit32.v:4	u_exe.u_alu_top.u_logic-logical_unit32	-
115	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_logic.rs1' of instance 'u_exe.u_alu_top.u_logic' is not a register"	logical_unit32.v:2	u_exe.u_alu_top.u_logic-logical_unit32	-
116	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_logic.rs2' of instance 'u_exe.u_alu_top.u_logic' is not a register"	logical_unit32.v:3	u_exe.u_alu_top.u_logic-logical_unit32	-
117	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_shift.alu_ctrl' of instance 'u_exe.u_alu_top.u_shift' is not a register"	shift_unit32.v:71	u_exe.u_alu_top.u_shift-shift_unit32	-
118	Waived (input should be register, but some inputs is taken from the	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_shift.rs1' of instance 'u_exe.u_alu_top.u_shift' is not a register"	shift_unit32.v:69	u_exe.u_alu_top.u_shift-shift_unit32	-

	combinational module so this warning can ignore)					
119	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_shift.rs2' of instance 'u_exe.u_alu_top.u_shift' is not a register"	shift_unit32.v:70	u_exe.u_alu_top.u_shift-shift_unit32	-
120	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_cmp.alu_ctrl' of instance 'u_exe.u_alu_top.u_cmp' is not a register"	compare_unit32.v:4	u_exe.u_alu_top.u_cmp-compare_unit32	-
121	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_cmp.rs_1' of instance 'u_exe.u_alu_top.u_cmp' is not a register"	compare_unit32.v:2	u_exe.u_alu_top.u_cmp-compare_unit32	-
122	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.u_cmp.rs_2' of instance 'u_exe.u_alu_top.u_cmp' is not a register"	compare_unit32.v:3	u_exe.u_alu_top.u_cmp-compare_unit32	-
123	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.alu_ctrl' of instance 'u_exe.u_alu_top' is not a register"	alu_top32.v:7	u_exe.u_alu_top-alu_top32	-
124	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.rs1' of instance 'u_exe.u_alu_top' is not a register"	alu_top32.v:5	u_exe.u_alu_top-alu_top32	-
125	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.u_alu_top.rs2' of instance 'u_exe.u_alu_top' is not a register"	alu_top32.v:6	u_exe.u_alu_top-alu_top32	-
126	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.alu_ctrl_ex' of instance 'u_exe' is not a register"	execute_stage.v:9	u_exe-execute_stage	-
127	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.data_forward_mem' of instance 'u_exe' is not a register"	execute_stage.v:13	u_exe-execute_stage	-
128	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.data_forward_wb' of instance 'u_exe' is not a register"	execute_stage.v:14	u_exe-execute_stage	-
129	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.ex_alu_src_ex' of instance 'u_exe' is not a register"	execute_stage.v:8	u_exe-execute_stage	-
130	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.imm_ex' of instance 'u_exe' is not a register"	execute_stage.v:4	u_exe-execute_stage	-
131	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.operand_a_forward_cntl' of instance 'u_exe' is not a register"	execute_stage.v:11	u_exe-execute_stage	-

132	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.operand_b_forward_cntl' of instance 'u_exe' is not a register"	execute_stage.v:12	u_exe-execute_stage	-
133	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.pc_ex' of instance 'u_exe' is not a register"	execute_stage.v:6	u_exe-execute_stage	-
134	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.rs1_data_ex' of instance 'u_exe' is not a register"	execute_stage.v:2	u_exe-execute_stage	-
135	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_exe.rs2_data_ex' of instance 'u_exe' is not a register"	execute_stage.v:3	u_exe-execute_stage	-
136	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.branch_ex' of instance 'u_branch' is not a register"	branch_jump_unit.v:5	u_branch-branch_jump_unit	-
137	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.carry_flag' of instance 'u_branch' is not a register"	branch_jump_unit.v:14	u_branch-branch_jump_unit	-
138	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.func3_ex' of instance 'u_branch' is not a register"	branch_jump_unit.v:8	u_branch-branch_jump_unit	-
139	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.imm_ex' of instance 'u_branch' is not a register"	branch_jump_unit.v:10	u_branch-branch_jump_unit	-
140	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.jal_ex' of instance 'u_branch' is not a register"	branch_jump_unit.v:6	u_branch-branch_jump_unit	-
141	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.jalr_ex' of instance 'u_branch' is not a register"	branch_jump_unit.v:7	u_branch-branch_jump_unit	-
142	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.negative_flag' of instance 'u_branch' is not a register"	branch_jump_unit.v:13	u_branch-branch_jump_unit	-
143	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.op1_forwarded' of instance 'u_branch' is not a register"	branch_jump_unit.v:16	u_branch-branch_jump_unit	-
144	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.overflow_flag' of instance 'u_branch' is not a register"	branch_jump_unit.v:15	u_branch-branch_jump_unit	-
145	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.pc_ex' of instance 'u_branch' is not a register"	branch_jump_unit.v:9	u_branch-branch_jump_unit	-

	this warning can ignore)					
146	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.predictedTaken_ex' of instance 'u_branch' is not a register"	branch_jump_unit.v:11	u_branch-branch_jump_unit	-
147	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_branch.zero_flag' of instance 'u_branch' is not a register"	branch_jump_unit.v:12	u_branch-branch_jump_unit	-
148	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_store.addr' of instance 'u_mem.u_datamem.u_store' is not a register"	store_datapath.v:40	u_mem.u_datamem.u_store-store_datapath	-
149	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_store.store_type' of instance 'u_mem.u_datamem.u_store' is not a register"	store_datapath.v:38	u_mem.u_datamem.u_store-store_datapath	-
150	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_store.write_data' of instance 'u_mem.u_datamem.u_store' is not a register"	store_datapath.v:39	u_mem.u_datamem.u_store-store_datapath	-
151	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_mem.addr' of instance 'u_mem.u_datamem.u_mem' is not a register"	data_memory.v:5	u_mem.u_datamem.u_mem-data_memory	-
152	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_mem.byte_enable' of instance 'u_mem.u_datamem.u_mem' is not a register"	data_memory.v:7	u_mem.u_datamem.u_mem-data_memory	-
153	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_mem.clk' of instance 'u_mem.u_datamem.u_mem' is not a register"	data_memory.v:2	u_mem.u_datamem.u_mem-data_memory	-
154	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_mem.mem_read' of instance 'u_mem.u_datamem.u_mem' is not a register"	data_memory.v:3	u_mem.u_datamem.u_mem-data_memory	-
155	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_mem.mem_write' of instance 'u_mem.u_datamem.u_mem' is not a register"	data_memory.v:4	u_mem.u_datamem.u_mem-data_memory	-
156	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_mem.write_data' of instance 'u_mem.u_datamem.u_mem' is not a register"	data_memory.v:6	u_mem.u_datamem.u_mem-data_memory	-
157	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_load.addr' of instance 'u_mem.u_datamem.u_load' is not a register"	load_datapath.v:9	u_mem.u_datamem.u_load-load_datapath	-
158	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_load.load_type' of instance 'u_mem.u_datamem.u_load' is not a register"	load_datapath.v:7	u_mem.u_datamem.u_load-load_datapath	-
159	Waived (input should be register, but some inputs is taken from the	INS_NR_INPR	"Input port 'u_mem.u_datamem.u_load.mem_data_in' of instance 'u_mem.u_datamem.u_load' is	load_datapath.v:8	u_mem.u_datamem.u_load-load_datapath	-

	combinational module so this warning can ignore)		not a register"			
160	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.addr' of instance 'u_mem.u_datamem' is not a register"	data_mem_top.v:7	u_mem.u_datamem-data_mem_top	-
161	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.clk' of instance 'u_mem.u_datamem' is not a register"	data_mem_top.v:2	u_mem.u_datamem-data_mem_top	-
162	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.load_type' of instance 'u_mem.u_datamem' is not a register"	data_mem_top.v:5	u_mem.u_datamem-data_mem_top	-
163	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.mem_read' of instance 'u_mem.u_datamem' is not a register"	data_mem_top.v:3	u_mem.u_datamem-data_mem_top	-
164	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.mem_write' of instance 'u_mem.u_datamem' is not a register"	data_mem_top.v:4	u_mem.u_datamem-data_mem_top	-
165	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.rs2_data' of instance 'u_mem.u_datamem' is not a register"	data_mem_top.v:8	u_mem.u_datamem-data_mem_top	-
166	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.u_datamem.store_type' of instance 'u_mem.u_datamem' is not a register"	data_mem_top.v:6	u_mem.u_datamem-data_mem_top	-
167	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.alu_result_mem' of instance 'u_mem' is not a register"	mem_stage.v:76	u_mem-mem_stage	-
168	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.clk' of instance 'u_mem' is not a register"	mem_stage.v:74	u_mem-mem_stage	-
169	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.mem_load_type_mem' of instance 'u_mem' is not a register"	mem_stage.v:82	u_mem-mem_stage	-
170	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.mem_store_type_mem' of instance 'u_mem' is not a register"	mem_stage.v:84	u_mem-mem_stage	-
171	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.mem_write_mem' of instance 'u_mem' is not a register"	mem_stage.v:80	u_mem-mem_stage	-
172	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.memtoreg_mem' of instance 'u_mem' is not a register"	mem_stage.v:86	u_mem-mem_stage	-

173	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_mem.rs2_data_mem' of instance 'u_mem' is not a register"	mem_stage.v:78	u_mem-mem_stage	-
174	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_wb.alu_result_wb' of instance 'u_wb' is not a register"	wb_stage.v:5	u_wb-wb_stage	-
175	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_wb.load_data_wb' of instance 'u_wb' is not a register"	wb_stage.v:6	u_wb-wb_stage	-
176	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_wb.memtoereg_wb' of instance 'u_wb' is not a register"	wb_stage.v:9	u_wb-wb_stage	-
177	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_wb.rd_wb' of instance 'u_wb' is not a register"	wb_stage.v:7	u_wb-wb_stage	-
178	Waived (input should be register, but some inputs is taken from the combinational module so this warning can ignore)	INS_NR_INPR	"Input port 'u_wb.wb_reg_file_wb' of instance 'u_wb' is not a register"	wb_stage.v:8	u_wb-wb_stage	-
179	Waived (having both synchronous and asynchronous logic, but writing to memory is synchronous and reading from memory is asynchronous.so this warning can be ignore. In modules such as BTB storage and dynamic branch predictor logic, combinational logic is used for same-cycle read paths and next-state computation, while sequential logic is used for state storage and updates. This separation is architecturally required to meet instruction fetch and branch prediction timing requirements. This mixed synchronous/asynchronous structure is intentional,synthesizable, and consistent with standard BTB and branch predictor implementations. No functional or timing risk exists.)	MOD_IS_SYAS	"The design-unit 'btb_file' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'rd_valid1' and Synchronous logic at 'valid_arr[0][1]'"	btb_file.v:44	*-btb_file	-
180	Waived (having both synchronous and asynchronous logic, but writing to memory is synchronous and reading from memory is asynchronous.so this warning can be ignore. In modules such as BTB storage and dynamic branch predictor logic,	MOD_IS_SYAS	"The design-unit 'dynamic_branch_predictor' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'state_n[1]' and Synchronous logic at 'next_state[1]'"	dynamic_branch_predictor.v:14	*-dynamic_branch_predictor	-

	combinational logic is used for same-cycle read paths and next-state computation, while sequential logic is used for state storage and updates. This separation is architecturally required to meet instruction fetch and branch prediction timing requirements. This mixed synchronous/asynchronous structure is intentional, synthesizable, and consistent with standard BTB and branch predictor implementations. No functional or timing risk exists.)					
181	Waived (having both synchronous and asynchronous logic, but writing to memory is synchronous and reading from memory is asynchronous. so this warning can be ignore. In modules such as BTB storage and dynamic branch predictor logic, combinational logic is used for same-cycle read paths and next-state computation, while sequential logic is used for state storage and updates. This separation is architecturally required to meet instruction fetch and branch prediction timing requirements. This mixed synchronous/asynchronous structure is intentional, synthesizable, and consistent with standard BTB and branch predictor implementations. No functional or timing risk exists.)	MOD_IS_SYAS	"The design-unit 'register_file' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'rs2_data[31]' and Synchronous logic at 'reg_file[1][31]'"	register_file.v:17	*-register_file	-
182	Waived (Top-level inputs are registered in upstream logic)	MOD_NO_IPRG	"Input port 'rst_im' of top-level design-unit is not registered"	rv32i_core.v:8	rv32i_core-rv32i_core	-
183	Waived (Top-level inputs are registered in upstream logic)	MOD_NO_IPRG	"Input port 'write_addr' of top-level design-unit is not registered"	rv32i_core.v:10	rv32i_core-rv32i_core	-
184	Waived (Top-level inputs are registered in upstream logic)	MOD_NO_IPRG	"Input port 'write_data' of top-level design-unit is not registered"	rv32i_core.v:11	rv32i_core-rv32i_core	-
185	Waived (Top-level inputs are registered in upstream logic)	MOD_NO_IPRG	"Input port 'write_en' of top-level design-unit is not registered"	rv32i_core.v:9	rv32i_core-rv32i_core	-
186	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:175	rv32i_core-rv32i_core	-
187	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:434	rv32i_core-rv32i_core	-

188	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:435	rv32i_core-rv32i_core	-
189	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:454	rv32i_core-rv32i_core	-
190	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:451	rv32i_core-rv32i_core	-
191	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:452	rv32i_core-rv32i_core	-
192	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:367	rv32i_core-rv32i_core	-
193	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:453	rv32i_core-rv32i_core	-
194	Waived (glue logic, but for synthesis we want some output ports so glue logic is used)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'rv32i_core'"	rv32i_core.v:455	rv32i_core-rv32i_core	-
195	Waived (Instruction and data memory arrays are initialized externally (HEX/mask ROM). Lint cannot infer these drivers; behavior is correct by design)	NET_NO_LDDR	"Net 'u_if.u_imem.mem' declared in design-unit 'inst_mem' neither has driver nor any load"	inst_mem.v:11	u_if.u_imem-inst_mem	-
196	Waived (Instruction and data memory arrays are initialized externally (HEX/mask ROM). Lint cannot infer these drivers; behavior is correct by design)	NET_NO_LDDR	"Net 'u_mem.u_datamem.u_mem.mem' declared in design-unit 'data_memory' neither has driver nor any load"	data_memory.v:13	u_mem.u_datamem.u_mem-data_memory	-

module rv32i_core

created date Feb 10 04:20:12 IST 2023

[JasperGold 2019.12] [Platform Linux] [Version 2019.12p002 64 bits] [Build date 2020.02.27 18:30:54 PST]